

Package Pickup Monitoring System

ASP.NET Core MVC Laboratory Exercise Specification

Objective

Develop an ASP.NET Core MVC application to monitor incoming packages and pickups using Model Binding, Cookie Authentication, Validation, Repository Pattern (in-memory), Razor Views, Bootstrap, CRUD, and Search.

Business Scenario

Reception staff record incoming packages, monitor unclaimed items, and record package claims for residents or employees.

Learning Objectives

MVC, Model Binding, DTOs, Validation, Cookie Authentication, Authorization, Repository Pattern, CRUD, Search, Bootstrap.

Functional Requirements

Register User, Login, Logout, Register Package, View Monitoring List, Edit Package, View Details, Mark Package Claimed, Search.

Package Fields

Tracking Number, Recipient Name, Unit/Office, Contact Number, Courier Company, Package Type, Arrival Date & Time, Expected Pickup Date, Received By, Notes.

Monitoring

Display Tracking Number, Recipient, Courier, Package Type, Arrival Time, Claimed Time, Status, Actions.

Status

Waiting for Pickup, Claimed.

Validation

Use Required, StringLength, Phone, Display, DataType and validate ModelState.

Authentication

Public: Login/Register. Protected: Package pages using Cookie Authentication and [Authorize].

Entities

User: Id, FirstName, LastName, Email, Username, Password. Package: Id, TrackingNumber, RecipientName, UnitNumber, ContactNumber, CourierCompany, PackageType, ArrivalDateTime, ExpectedPickupDate, ClaimedDateTime, ReceivedBy, Status, Notes.

Repositories

UserRepository and PackageRepository using static List.

Controllers & Views

Controllers: HomeController, AccountController, PackageController. Views: Login, Register, Index, Create, Edit, Details, Claim.

Submission

Complete MVC application with authentication, CRUD, validation, search, Bootstrap UI, repository pattern, and clean code.